

Speech and Video Processing (CS30033)

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INSTRUCTORS:

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Module I

1. Outline the high-level architecture of a modern Automatic Speech Recognition (ASR) system.
2. Describe the challenges of ASR due to background noise and room reverberation.
3. Explain the stages of human speech generation with a schematic diagram.
4. Explain the source-filter model of speech production. How do the vocal cords and the vocal tract act independently in shaping speech sounds?
5. What is the difference between voiced and voiceless phonemes? Provide examples.
6. The pronunciation of words differs when spoken in isolation compared to when spoken within continuous sentences. What challenges does this variability pose for Automatic Speech Recognition systems?
7. Explain the psychoacoustic principles of the Mel frequency scale. Why does a linear frequency representation fail to accurately capture the characteristics of human auditory perception?
8. What is the difference between voiced, unvoiced and silence regions in the context of the human speech production system? (b) How are voiced and unvoiced sounds produced?
9. What is the fundamental frequency (F_0) of speech? What factors determine the fundamental frequency of a speaker's voice?
10. Explain the production of plosive, fricative, nasal, and approximant sounds.

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11. What are F1, F2, and F3 formants? Why is F1 generally related to vowel openness?
Why is F2 useful for distinguishing front and back vowels?
 12. What is a speech signal? Is speech a periodic or aperiodic signal?
 13. Why is speech considered a non-stationary signal?
 14. What is meant by quasi-stationarity of speech?
 15. What is the typical frequency range of human speech? What is the sampling frequency required for speech processing?
 16. Why is 8 kHz commonly used in telephone speech?
 17. Why is 16 kHz commonly used in speech-processing applications?
 18. What is speech amplitude? What is speech intensity?
 19. What are poles and zeros of a transfer function? How do you obtain poles and zeros from a transfer function?
 20. How do poles and zeros affect system stability? What does the pole-zero plot represent?
 21. How do poles and zeros affect the frequency response?
 22. What is the significance of poles on the left/right half of the $j\omega$ -axis?
 23. What happens when a pole lies on the $j\omega$ -axis?
 24. What is pole-zero cancellation?
 25. Explain the reasoning behind the 44.1 KHz sampling frequency microphone to capture the full spectrum of human hearing.
 26. Explain the phenomenon of aliasing in the context of speech signals.
 27. What is the impulse response of a system? Discuss the different categories of filters based on impulse response duration.
 28. Explain the concept of pole-zero modeling in speech systems. How do Poles model formants in the Z-Domain?
 29. Discuss the criteria for pole placement within the Z-domain to ensure the stability of a digital speech synthesis filter.
 30. What is the time-domain representation of a speech signal?
 31. What is Zero-Crossing Rate (ZCR)?

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32. What are voiced and unvoiced segments in a waveform? How can ZCR be used to distinguish voiced and unvoiced speech?
 33. What is short-time energy? How is short-time energy calculated?
 34. What is an amplitude envelope? What is a spectral envelope?
 35. How can silence regions be detected from a speech waveform?
 36. Why is short-time analysis necessary for speech signals?
 37. Discuss the limitations of representing signals in the time domain and explain the advantages of analysing signals in the frequency domain.
 38. Explain spectral leakage and its impact on FFT-based spectral analysis.
 39. Write a short note on the role of twiddle factor in FFT. Estimate the computational complexity of computing the DFT versus using the FFT for $N=1024$. State the efficiency gain.
 40. What will be the Fourier transform of a discrete-time unit impulse signal? Show steps.
 41. Show steps of how Fast Fourier Transform (FFT) reduces complexity of Discrete Fourier transform. Why is stationary Fourier analysis insufficient for speech signals?
 42. State how human pitch perception is incorporated in filter-bank modelling for speech signal spectral analysis.
 43. Explain how the time-domain convolution of the glottal excitation source and the vocal tract impulse response translates into the frequency domain.
 44. Define speech processing. Write the necessity of speech processing in modern world applications.
 45. Distinguish between speech transmission and speech processing.
 46. List out some application areas of digital speech processing.
 47. Write the importance of velum in speech production systems.
 48. Identify different types of excitation source.
 49. Define phoneme. Identify different types of phonemes.
 50. Define vowels. Write down different types of vowels.
 51. Write the significance of F1 and F2 formant positions for vowels.

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52. Define F1-F2 Cetroid. What purpose it serves in the acoustic representation of speech sounds.
 53. Define F1-F2 Cluster. What purpose it serves in the acoustic representation of speech sound.
 54. Define diphthongs. How semivowels are different from diphthongs.
 55. Define consonants. List out different types of consonants.
 56. How is the syllable different from phoneme?
 57. Identify the number of syllables in the word "spectrum".
 58. Define amplitude, frequency, and phase for a sinusoidal signal.
 59. Differentiate between continuous time and discrete time signal.
 60. "A speech signal considered as non-stationary and quasi-periodic in nature." Write the significance of the above statement.
 61. Define digitization. Write the purpose of analog-to-digital converter in digitization.
 62. Define sampling frequency. How is it related to the sampling interval?
 63. For a given signal, if the sampling frequency is 16KHz then determine the sampling interval.
 64. Why sampling rate must not be below twice the highest frequency present in the signal.
 65. How is the nyquist rate related to the nyquist frequency?
 66. Define aliasing. Write the effect of aliasing in signal processing.
 67. List out the ways to avoid aliasing in signal processing.
 68. Define quantization. Why is it a significant step in signal processing?
 69. Define quantization step size.
 70. Define quantization error. How quantization step size affects mean squared quantization error.
 71. Discuss the time-domain representation of a speech signal.
 72. What information can be gathered from the time-domain representation of a speech signal?
 73. Discuss the frequency-domain representation of a speech signal.

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74. What information can be gathered from the time-domain representation of a speech signal?
 75. Write the significance of spectrograms in speech processing.
 76. List out the application areas where spectrograms can be useful.
 77. Define a linear time invariant system along with its mathematical representation.
 78. Explain the major properties that are necessary for a discrete-time system to be an LTI system.
 79. Write the principle of superposition with reference to an LTI system.
 80. Define convolution. What role does impulse response play in convolution?
 81. List out at least three properties of convolution.
 82. Briefly describe the finite duration property of convolution. (Prepare in a similar way for all the properties of convolution.)
 83. Write the shortcomings of time-domain convolution.
 84. Define pole-zero modeling.
 85. Why pole-zero modeling is considered as one of the important processes in speech processing.
 86. Describe vocal tract in terms of a transfer function in pole zero modeling.
 87. What poles and zeros represent in the pole-zero modeling?
 88. What will be the impact on a linear system if the poles are set to zero?
 89. What will be the impact on a linear system if the zeros are set to zero?
 90. What role does Fourier transform play in the speech processing?
 91. How can a sinusoidal signal be represented in complex exponential?
 92. Write about the importance of dirac delta function in Fourier transform.
 93. List out the properties of fourier transform.
 94. Define inverse fourier transform.
 95. Define discrete fourier transform along with its mathematical representation.
Define inverse DTFT.
 96. What is the basis function in DFT?
 97. Write the orthogonality principle of DFT basis function.

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98. Explain the spectral leakage. How can it be avoided?
 99. Discuss the significance of even odd decomposition in fast fourier transform.
 100. Define twiddle factor in FFT. Explain butterfly networks in FFT.
 101. What information does the spectral envelope carry?
 102. List out the different types of filters in signal processing.
 103. Distinguish between uniform and non-uniform digital filters.
 104. How is a full wave rectifier different from a half wave rectifier?
 105. Write the significance of Mel-Scale in speech processing.
 106. What are the shortcomings of the filter bank model?
 107. Define linear prediction along with its mathematical representation.
 108. What is prediction error in linear prediction?
 109. How does the prediction order impact a LPC model?
 110. What role does an all-pole filter play in the LPC model?
 111. Define the term excitation gain in LPC.
 112. List out the typical model parameters in LPC.
 113. Explain autocorrelation along with its mathematical representation.
 114. How does the covariance method differ from the autocorrelation method?
 115. Write the whitening behavior of the LPC model.
 116. Define vowels. Explain different types of vowels with suitable examples for each of them.
 117. How are the diphthongs and semi vowels different from each other? List out at least two examples for diphthongs and semi vowels.
 118. Explain consonants. Discuss different types of consonants with suitable examples.
 119. Compare and contrast the continuous-time and discrete-time signals. How is the continuous-valued signal different from the discrete-valued signal?
 120. Discuss the architecture of a digital signal processing system with a neat diagram.

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121. Define sampling. How does sampling interval and sampling frequency relate to each other? Write the significance of bandwidth in sampling.
 122. Explain Nyquist-Shannon sampling theorem. Discuss the issue of aliasing with respect to sampling. How can the aliasing be overcome?
 123. Define quantization step size along with its mathematical representation. How does the bits per sample affect the quantization step size?
 124. Explain the Fast Fourier Transform (FFT) algorithm used for efficient computation of the Discrete Fourier Transform (DFT). Describe the principle of divide-and-conquer and explain how the computational complexity of FFT is reduced compared to direct DFT computation.
 125. Explain the complete filter bank model structure used for spectral estimation of speech signals with a suitable block diagram.
 126. Describe the basic structure of a filter bank and explain how a bank of bandpass filters helps in representing the spectral characteristics of a speech signal.
 127. Explain the principle of Linear Predictive Coding (LPC) in speech processing. Describe how the current speech sample is predicted from previous samples and explain the significance of LPC coefficients, prediction error, and prediction order in representing the speech signal.
 128. Explain the major steps involved in LPC analysis of a speech signal. Describe the role of pre-emphasis, framing and windowing, autocorrelation, LPC coefficient estimation, and parameter conversion.
 129. Explain the autocorrelation method for estimating LPC coefficients. How is the autocorrelation sequence used to formulate the LPC normal equations? Mention one advantage of the autocorrelation method.
 130. Differentiate between the autocorrelation method and covariance method of LPC analysis with respect to the analysis interval, treatment of boundary samples, and prediction-error minimization.

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131. The prediction order p is an important parameter in LPC analysis. Explain what may happen if the prediction order is chosen too low or too high. How would you select a suitable prediction order for speech signals?
 132. What is prediction error in Linear Predictive Coding? Explain how minimizing the prediction error helps in estimating the LPC coefficients and how the resulting error signal is related to the excitation of the speech production system.
 133. Discuss the time-domain and frequency-domain representations of a speech signal. Write the significance of the axes used in both the representations.
 134. Differentiate between time-domain and frequency domain representations of a speech signal.
 135. Define spectrograms. How can a spectrogram be created for a given speech signal? Why is the spectrogram needed in the speech signal processing?
 136. Define Linear Time Invariant (LTI) system. Discuss the necessary properties for a discrete system to be an LTI system.

Module II

1. What is cepstral Analysis of speech signals and why is it required?
2. Explain the steps required to find the cepstrum of a signal.
3. What are the similarities and differences between real and complex cepstrums?
4. "The vocal tract changes slowly and its log-spectrum is smooth". True or false? Justify your answer using the concept of cepstral analysis.
5. Note the difference between static and dynamic features of speech.
6. Explain different types of static features.
7. How are temporal features different from spectral features?
8. Why are cepstral features preferred in speech applications?
9. How prosodic features are different from Glottal features?
10. "MFCC represents the power spectrum of an audio signal based on how the human ear actually perceives sound". True or False. Justify your answer.
11. Why is pre-emphasis carried out during extraction of MFCC?
12. What is frame blocking? How is it different from windowing?
13. What is the requirement of using Mel Filter bank during MFCC extraction?
14. Why are overlapping windows used instead of clear cut windows?
15. What are the most commonly used windows in MFCC?
16. How many coefficients are produced during MFCC feature extraction?
17. Why is it assumed that 12 to 13 coefficients are produced in MFCC?
18. With a suitable diagram, explain the process of MFCC feature extraction.
19. What are applications where MFCC features are preferred over LPCC?
20. What coefficients are there in the LPC parameter set?
21. "LPCC are cepstral coefficients and appended vectors of cepstral time derivative"
True or False?
22. With a suitable diagram, explain the process of MFCC feature extraction.
23. What is gained by transforming a convolution to an addition with the cepstrum?

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24. Why is taking the inverse Fourier transform the same as taking the Fourier transform when computing the cepstrum?
 25. Following framing and windowing, the Fast Fourier Transform (FFT) is applied, and the magnitude spectrum is squared to yield the power spectrum. Why is phase information discarded at this stage?
 26. Why is the Discrete Cosine Transform (DCT) applied as the final step in standard MFCC feature extraction over the log-mel energies?
 27. In automatic speech recognition (ASR) implementations, higher-order cepstral coefficients are often discarded, and a "liftering" operation is applied. What is the primary operational effect of liftering?
 28. While MFCCs are foundational in speech recognition, what is a well-documented vulnerability of standard MFCCs in noisy or reverberant real-world environments?
 29. What is a codebook in Vector Quantization, and how does it achieve data compression by mapping continuous vectors to discrete code vectors?
 30. What are the major Pattern Matching approaches for ASR?
 31. Briefly explain the three basic problems of HMMs—Evaluation (Forward-Backward), Decoding (Viterbi), and Learning (Baum-Welch).

Numerical Problems:

1. A speech signal has a maximum frequency of 3 kHz. It is sampled at 8 kHz using 16-bit uniform quantization. Find the Nyquist rate and check if the sampling frequency is sufficient. Find the quantization step size (Δ) and maximum quantization error if the signal amplitude range is ± 2 V.

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2. Consider a system defined as: $y[n] = x[n] \cdot (-1)^n$ With the input sequence:
 $x[n] = \{1, 7, 3\}$ for $n = 0, 1, 2$ and shift (n_0) = 1, verify numerically whether this system is time-invariant.
 3. Input signal: $x[n] = \{4, 1, 3, 4, 2\}$ at $n = 0, 1, 2, 3, 4$. Impulse response:
 $h = \{1, -1, 0, 1\}$. Apply convolution.
 4. A continuous-time signal $x(t) = \cos(2000\pi t)$ is sampled at a rate of $f_s = 2500$ Hz to produce the discrete-time sequence $x[n]$. Find the discrete-time frequency ω (in rad/sample) of the sampled sequence.
 5. The z-transform of an LTI system is

$$H(z) = \frac{1}{1 - 0.5 z^{-1}}, \quad |z| > 0.5$$

Find the poles and zeroes for the system.
 6. Find the poles and zeros of $H(s) = (s+2)/(s^2+5s+6)$
 7. Sketch the pole-zero plot for $H(s) = (s+1)(s+4)/(s+2)(s^2+4s+13)$
 8. Determine whether the system is stable: $H(s) = 10/(s+1)(s+3)(s-2)$
 9. Find the transfer function from the following poles and zeros: Zeros: $s=-2, -5$ Poles: $s=-1, -3, -4$ Gain: $k=10$
 10. Determine poles and zeros and stability for system with $H(s) = (s+1)/(s^2+2s+10)$
 11. A discrete-time signal is given as: $x[n] = \{3, -1, 2, -2\}$, $n = 0, 1, 2, 3$

Compute the 4-point DFT of $x[n]$. Show all steps clearly.
 12. A Mel-scale filter bank is designed for an audio signal sampled at $f_s = 16$ kHz, the frequency ranges from 0 Hz to the Nyquist frequency (8 kHz). If you want to construct a filter bank with $M = 4$ Mel filters that are evenly spaced on the Mel scale, what will be the center points (Mel scale vs. Hz) for the banks?
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13. A speech signal is uniformly quantized using 8 bits per sample over a range of ± 2 V. If the sampling frequency is 16 kHz, determine the quantization step size (Δ). Also, find out the size of the file for a 5-second recording
14. A speech signal has an amplitude range of -1.5 V to $+1.5$ V and is quantized using 10 bits per sample. The sampling frequency is 16 kHz. For a recording duration of 20 seconds, determine:
- The quantization step size.
 - The total number of samples.
 - The total number of bits required.
 - The approximate file size in KB
15. A 10-bit ADC converts analog signals in the range -5 V to $+5$ V. Determine Quantization step size, maximum quantization error, and mean squared quantization error.
16. A speech signal is uniformly quantized using 6 bits per sample over a range of ± 3 V. Find the quantization step size (Δ).
17. Suppose the ADC is modified to operate with a 12-bit resolution instead of 10-bit resolution. Discuss how this change affects the ADC's operation.
18. A speech acquisition system samples a signal at 16 kHz using a 10-bit ADC over the input range -1.5 V to $+1.5$ V. Calculate
- Quantization step size
 - Maximum quantization error
 - Mean squared quantization error
19. Consider the following transformation: $y[n] = x[n] + 2x[n-2]$ For the input sequence $x[n] = \{2, 4, 6\}$, Check whether the given system is time-invariant or time-varying.

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20. A discrete-time LTI system has the input signal $x[n] = \{2, 1, 2, 4, 3\}$, $n=0, 1, 2, 3, 4$; and the impulse response $h[n]=\{1, -1, 2\}$, $n=0, 1, 2$. Using the convolution operation, determine the output sequence $y[n]$.
21. A discrete-time LTI system has the input signal $x[n]=\{1, 3, 2, 1\}$, $n=0, 1, 2, 3$ and the impulse response $h[n]=\{2, -1\}$, $n=0, 1$. Using the convolution operation, determine the output sequence $y[n]$.
22. A discrete-time LTI system has the input signal $x[n] = \{1, 2, 1, 3\}$, $n = 0, 1, 2, 3$ and the impulse response $h[n] = \{1, 0, -1\}$, $n = 0, 1, 2$. Using the convolution operation, determine the output sequence $y[n]$.
23. Express $x[n]=\{3, 4, 5\}$ as a sum of impulses and use convolution with $h[n]=\{1, 2\}$ to find $y[n]$.
24. $x[n]=\{2, -1, 3\}$, $h[n]=\{1, -2, 1\}$. Compute $y[n]$ and verify commutative property.
25. $x[n]=\{1, 2, 1, 3\}$, $h[n]=\{1, 0, -1\}$. Determine output length and compute all $y[n]$.
26. Discuss the limitations of time-domain convolution.
27. Explain the concept of pole-zero modeling in speech processing. What is the significance of poles and zeros in representing the characteristics of the vocal tract?
28. Find the poles and zeros of $H(z) = \frac{z^2(z+1)}{(z-0.8)(z+0.5)}$
29. Determine poles and zeros for $H(z) = \frac{(z+0.5+j0.5)(z+0.5-j0.5)}{(z-0.6+j0.3)(z-0.6-j0.3)}$
30. For the given signals, find the frequency of each cosine, exponential representation and Fourier Transform:
- $x(t)=6\cos(120\pi t)+4\cos(40\pi t)$
 - $x(t)=4\cos(2\pi 20t)+2\cos(2\pi 60t)$
 - $x(t)=3\cos(2\pi 15t)+5\cos(2\pi 50t)$
 - $x(t)=8\cos(2\pi 10t)+4\cos(2\pi 25t)+6\cos(2\pi 40t)$
31. Suppose a sine wave $x(t)=2\sin(2\pi ft)$ with 1 Hz frequency, amplitude 2, sampling frequency: 8 Hz. Apply 8-point DFT.
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32. Suppose a sine wave $x(t)=3\sin(2\pi ft)$ with 1 Hz frequency, amplitude 3, sampling frequency: 16 Hz. Apply 8-point DFT.
33. A sine wave has: Frequency: $f_0=2$ Hz, Amplitude: $A=1$ Sampling frequency: $F_s =16$ Hz. Apply an 8-point DFT to the sampled signal.
34. A speech processing system uses a Mel-scale filter bank with 20 filters. The frequency range is from 300 Hz to 8000 Hz. Calculate the mel-scale spacing between adjacent filters. Also, determine the corresponding Mel-scale frequencies for the filters.